



SCHOOL OF COMPUTER SCIENCE & IT
Beaconhouse National University

(Multivariable Calculus - B)

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Quiz no 3

STUDENT NAME

ROLL NUMBER

Maximum Marks: 5

Time: 15:00 minutes

Questions (1 × 5 = 5)

1. Consider the function:

$$f(x, y) = \sum_{n=0}^{\infty} (xy)^n, \quad \text{where } |xy| < 1.$$

- (a) Simplify the given summation to its closed form using the geometric series formula.
- (b) Find the first-order partial derivatives f_x and f_y .
- (c) Compute the second-order partial derivatives f_{xx} , f_{yy}